

Clustering Countries Based on AI Workforce Readiness and Automation Indicators Using K-Means

Walton Karl Avillanosa¹, John Romson Erazo¹, Claire Dane Vincoy¹, and Ara Abigail Ambita¹

University of the Philippines Visayas
wlavillanosa@up.edu.ph, jeerazo@up.edu.ph, cdvincoy@up.edu.ph

Abstract. This study applies K-Means clustering to group countries based on artificial intelligence workforce readiness and automation-related indicators from 2015 to 2025. The dataset includes indicators such as AI investment, automation rate, employment rate, productivity index, reskilling investment, AI policy index, job displacement, job creation, and AI readiness score. After preprocessing and standardization, a five-cluster solution was identified through elbow analysis, silhouette score evaluation, and cluster interpretability. The resulting clusters revealed distinct country profiles, including a high-investment, high-readiness group and an automation-heavy transition with low reskilling support group. The findings suggest that AI workforce readiness is shaped more by investment and policy decisions than by regional proximity. It also suggests that countries vary considerably in their approaches to managing automation-related workforce transitions.

Keywords: K-Means clustering · Artificial intelligence · Workforce readiness · Automation · Un-supervised learning

1 Introduction

Artificial intelligence (AI) has become one of the rapidly growing technological developments in the world, particularly in the areas of work and employment [3]. AI refers to human-like intelligence embedded in systems, enabling the automation of tasks that were traditionally performed by humans [4]. Although these technologies existed before 2020, the global pandemic accelerated their adoption. Because of that, AI became widely recognizable in industries and everyday life [5]. Since then, industries such as manufacturing, finance, healthcare, and logistics have undergone several AI-driven transformations.

Artificial intelligence and automation are increasingly affecting workforce structures, productivity, employment, and national development strategies. Padmaja et al. [3] highlighted that automation may replace some routine tasks while also creating demand for new technical and analytical roles. Similarly, Fedyk et al. [6] found that countries with stronger AI investment tend to experience productivity growth and increased employment, however, the distribution across skill levels remains uneven. Countries differ substantially in how they invest in AI and adopt automation for AI-driven labor market changes. Workforce reskilling and policy readiness are therefore important factors in adapting to these changes [13,12]. However, the lack of consistent global classifications for AI workforce readiness makes it difficult to compare countries or identify shared challenges. Because indicators vary across countries, grouping nations with similar AI workforce readiness and automation profiles becomes a useful and practical approach for analysis. Clustering methods such as K-Means are used in labor market studies to group economies with similar economic and employment characteristics when there are no predefined categories available [7]. This makes it useful for analyzing diverse country data.

This project aims to cluster countries based on AI workforce readiness and automation-related indicators using K-Means clustering.^{1 2} The dataset contains country-level AI workforce and automation indicators from 2015 to 2025, including AI investment, automation rate, employment rate, productivity index, reskilling investment, AI policy index, job displacement, job creation, and AI readiness score.

The objective of this study is to discover meaningful country groupings based on these indicators. Since the dataset has no predefined class labels, an unsupervised learning approach is used. The project expects that countries with similar AI investment, automation adoption, workforce readiness, and labor market outcomes will be grouped into the same clusters. Furthermore, it is expected that countries with stronger investment, productivity, policy readiness, or workforce adaptation indicators may form distinct

¹ GitHub repository: <https://github.com/cdvincoy/Mini-Project-197>

² Dataset source: <https://www.kaggle.com/datasets/emirhanakku/ai-workforce-and-automation-dataset-20152025/data>

clusters from countries with lower readiness or higher labor-market disruption, reflecting structural differences in how nations are positioned to absorb and benefit from AI-driven transformation. A limitation of this study is the exclusive use of K-Means clustering. Other methods such as hierarchical clustering or DBSCAN could reveal different patterns in the data. Future work may apply these methods to further validate and improve the results.

2 Methodology

2.1 Dataset

This study uses the AI Workforce and Automation Dataset sourced from Kaggle, which covers 20 countries from 2015 to 2025. The raw dataset, `global_ai_workforce_automation_2015_2025.csv`³, contains 220 rows and 12 columns, with no missing values across any variable. The 20 countries represented are: Australia, Brazil, Canada, China, France, Germany, India, Italy, Japan, Mexico, Netherlands, Russia, Singapore, South Africa, South Korea, Spain, Sweden, Turkey, United Kingdom, and United States.

Because the dataset contains yearly records and the goal is country-level analysis, records were aggregated by computing the mean value of each numeric feature per country. This produced a final dataset of 20 country-level observations, each representing a country’s average AI workforce across the 2015 to 2025 period.

2.2 Descriptive Analytics

Prior to modeling, a statistical summary was computed on the post-aggregated, pre-scaled values. Table 1 presents the mean, standard deviation, minimum, quartiles, and maximum for each of the features, excluding Year and Country.

Feature	Mean	Std. Dev.	Min	25%	Max
AI_Investment_BillionUSD	82.36	37.53	10.83	52.16	127.78
Automation_Rate_Percent	27.35	8.44	15.23	21.53	43.48
Employment_Rate_Percent	68.82	5.84	59.78	64.02	78.24
Productivity_Index	60.61	10.41	46.71	52.60	78.42
Reskilling_Investment_MillionUSD	939.96	450.76	247.17	573.70	1660.08
AI_Policy_Index	0.60	0.05	0.47	0.58	0.68
Job_Displacement_Million	0.37	0.10	0.21	0.27	0.54
Job_Creation_Million	1.25	0.59	0.21	0.87	2.27
AI_Readiness_Score	43.92	3.42	37.10	41.57	51.43
Average_Salary_USD	62074.48	18649.23	30396.57	47370.77	98701.36

Table 1. Descriptive statistics of selected features using country-level averages.

The statistics reveal considerable spread across all features. AI investment and reskilling investment, and average salary displayed the highest variation, suggesting substantial differences among the 20 countries in terms of financial commitment to AI and workforce development. Next, automation rate and productivity index displayed moderate variation. Meanwhile, AI policy index, job displacement, and job creation displayed relatively low variation, indicating that countries are more similar in these areas.

Figure 1 shows the correlation heatmap of the features from the dataset post-aggregation. Most features show moderate correlations with one another. The strongest positive correlations are observed between AI Investment and Job Creation (0.95) and Job Displacement and Automation Rate (0.90).

Figure 2 shows the distribution of each feature from the dataset after aggregation. Several features displayed wide spread, indicating substantial differences among the countries. Meanwhile, features such as AI Policy Index and AI Readiness show narrower distributions, suggesting similarities among the countries.

³ File: https://github.com/cdvincoy/Mini-Project-197/blob/main/global_ai_workforce_automation_2015_2025.csv

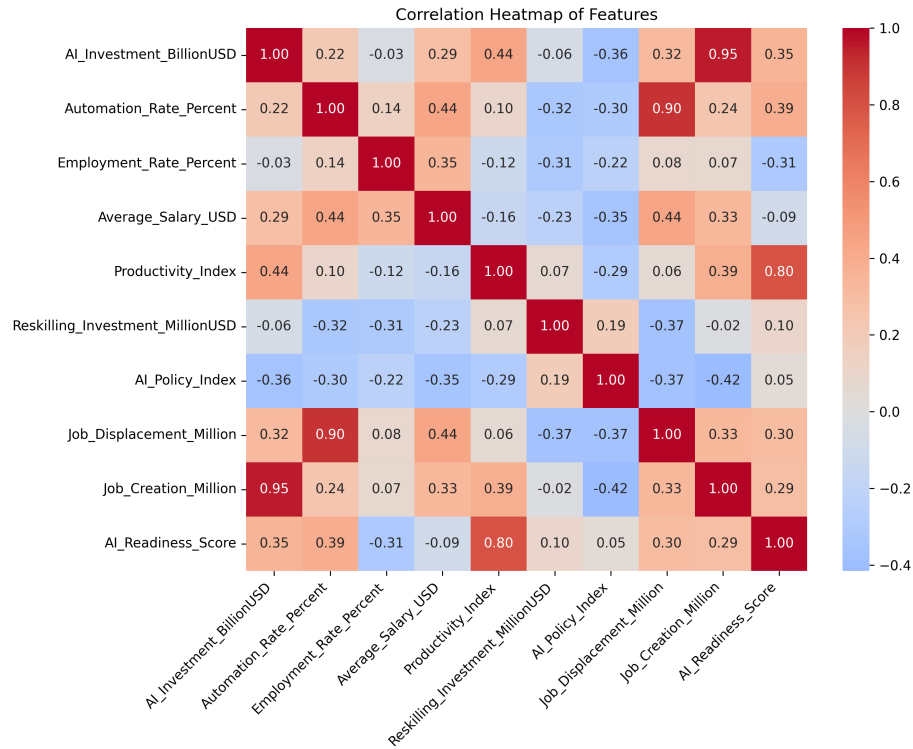


Fig. 1. Correlation heatmap of the dataset.

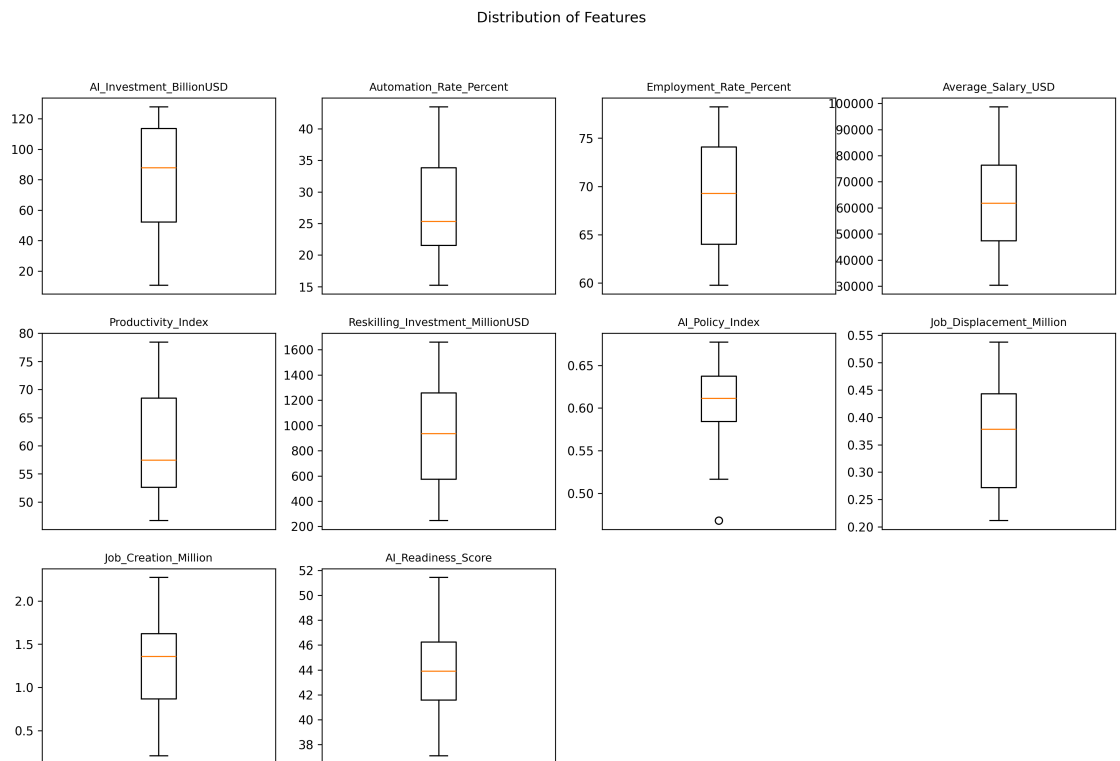


Fig. 2. Distribution of features from the dataset.

2.3 Feature Selection and Description

Out of the 12 available variables, nine features were selected for clustering. Year and country were not included in the selection. The `Average.Salary.USD` variable was excluded because it primarily reflects general economic development rather than AI-specific workforce readiness. The nine retained features are organized into three conceptual groups in Table 2.

Feature	Category	Description
<code>AI_Investment_BillionUSD</code>	Investment & Capacity	Annual AI investment in billions of USD.
<code>Reskilling_Investment_MillionUSD</code>	Investment & Capacity	Investment in workforce reskilling programs, measured in millions of USD.
<code>Employment_Rate_Percent</code>	Workforce & Labor Outcomes	Percentage of the working-age population that is employed.
<code>Job_Displacement_Million</code>	Workforce & Labor Outcomes	Estimated number of jobs displaced by automation, measured in millions.
<code>Job_Creation_Million</code>	Workforce & Labor Outcomes	Estimated number of new AI-driven jobs created, measured in millions.
<code>Productivity_Index</code>	Readiness & Policy	Workforce productivity score on a scale of 0 to 100.
<code>Automation_Rate_Percent</code>	Readiness & Policy	Percentage of the workforce subject to automation.
<code>AI_Policy_Index</code>	Readiness & Policy	National AI policy readiness index, measured from 0 to 1.
<code>AI_Readiness_Score</code>	Readiness & Policy	Composite readiness and AI adoption index.

Table 2. Selected features and their descriptions.

2.4 Data Preprocessing

As previously discussed, there were no missing values from the dataset. Moreover, since the dataset contains yearly records for each country, the data was grouped by country. In this study, the clustering model analyzed each country’s average AI workforce and automation profile across the 2015–2025 period.

After feature selection, standardization was applied using `StandardScaler`. This step was necessary because the features had different units and numerical ranges. For example, AI policy index is measured on a small scale, while reskilling investment is measured in millions of USD. Standardization prevented features with larger numerical values from dominating the K-Means distance calculations.

2.5 Clustering Model: K-Means

K-Means clustering was used because the project is an unsupervised learning task [7]. The algorithm groups data points by minimizing the distance between each point and the centroid of its assigned cluster. In this study, each data point represents one country after aggregation.

The implementation used the K-Means function from Scikit-learn with a fixed random state of 42 for reproducibility and `n_init=10`, which means the algorithm was run 10 times with different random centroid initializations and the solution with the lowest inertia was retained.

2.6 Determining the Number of Clusters

Selecting the appropriate number of clusters k is a critical step that affects the quality and under-standability of the results. Two methods were used to evaluate candidate values of k ranging from 2 to 10.

The elbow method was used to observe the decrease in inertia as the number of clusters increased. The objective is to identify the point at which additional clusters produce diminishing reductions, which form an elbow in the curve.

Silhouette scores were computed to evaluate how well-separated the clusters were [8]. It provides a quantitative measure of cluster quality. For each observation, the silhouette value is calculated based on how similar it is to its own cluster compared to the nearest neighboring cluster. Scores range from -1 to +1, where values close to +1 indicate that the observation is well-matched to its own cluster and clearly separated from others. Values near 0 suggest the observation lies on or near the boundary between clusters. The overall silhouette score is the mean across all observations.

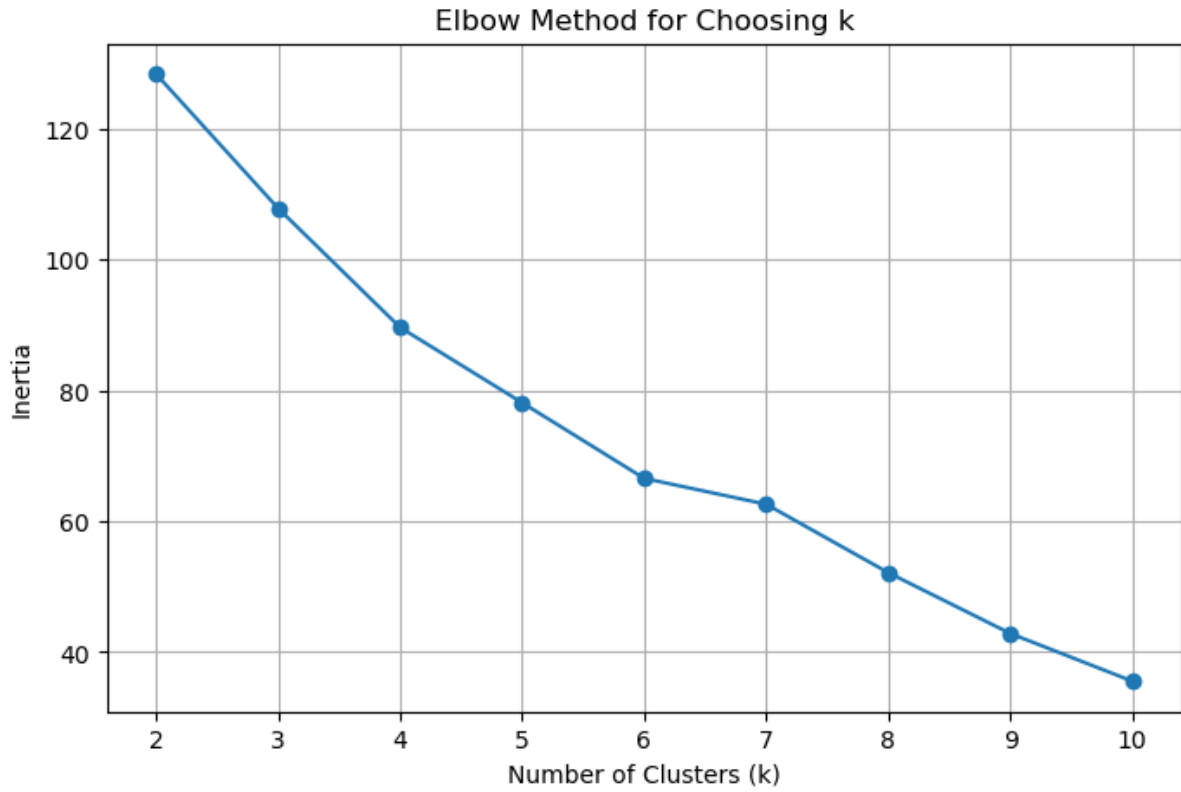


Fig. 3. Elbow method showing the decrease in inertia for different numbers of clusters.

Figure 3 shows that inertia decreases as the number of clusters increases. The decrease is expected because additional clusters allow data points to be closer to their assigned centroids. However, the elbow method alone does not fully determine the best value of k , so silhouette scores were also considered.

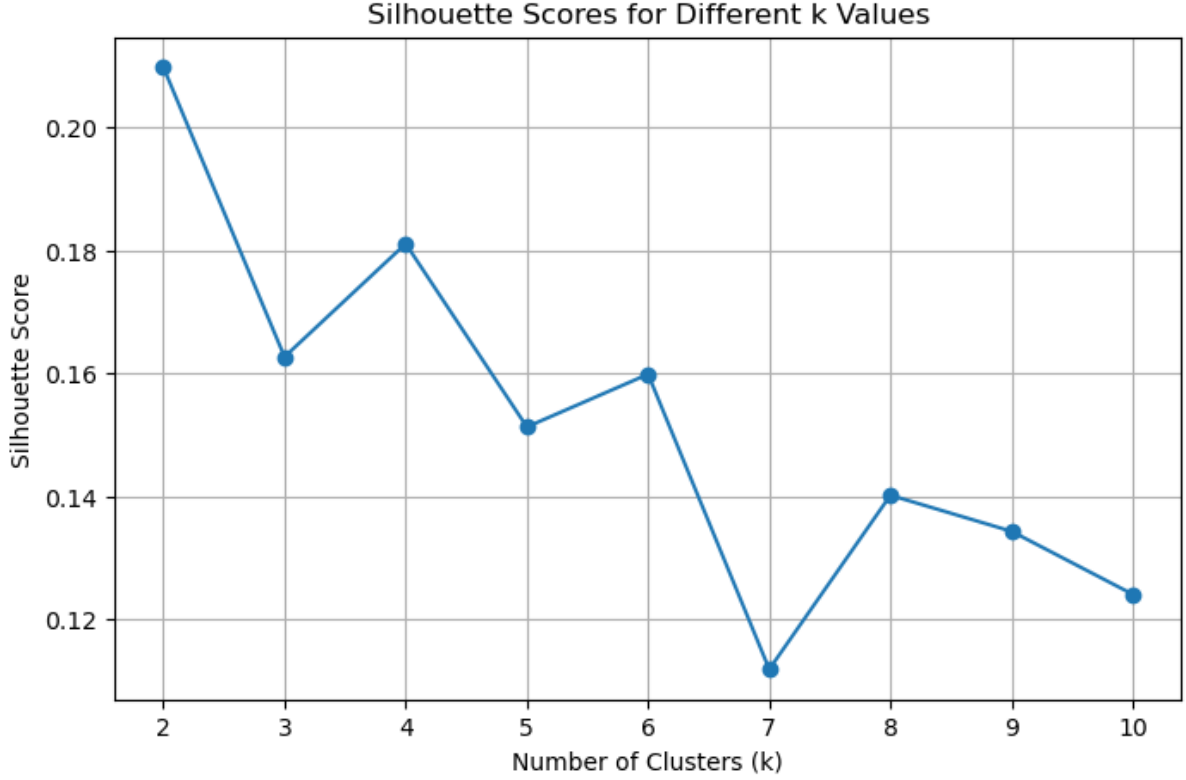


Fig. 4. Silhouette score comparison across different values of k .

Figure 4 shows the silhouette score comparison across different values of k . Although smaller values of k , in this case $k=2$, produced higher silhouette scores, the $k=5$ was selected as it provided more granular and interpretable country groupings relevant to the research objectives, while maintaining an acceptable silhouette score.

2.7 Visualization Methods

To visualize the cluster structure beyond the nine-dimensional feature space, Principal Component Analysis (PCA) was applied to the standardized data, reducing it to two components for a two-dimensional scatter plot. PCA identifies the directions of maximum variance in the data and projects observations onto a lower-dimensional subspace [9][9]. It is important to note that PCA was used exclusively for visualization. The K-Means clustering itself was performed in the full nine-dimensional feature space.

An interactive choropleth world map was also generated using Plotly Express's choropleth function to complement the scatter plot. This map displays each country's cluster assignment geographically, allowing for more intuitive understanding of regional patterns and groupings across the globe.

2.8 Performance Metrics

Cluster validation was used to evaluate the quality of the K-Means clustering results. Three metrics were computed to assess both cohesion and separation among clusters. The Silhouette Index was calculated for each country individually as well as overall, measuring how similar each observation is to its own cluster compared to neighboring clusters. Values that are closer to +1 indicate well-separated and cohesive clusters. Davies-Bouldin Index was computed overall. The index evaluates clustering quality by comparing within-cluster dispersion to between-cluster separation. [10] Lower values indicate better cluster separation. Finally, Dunn's Index was also computed overall, where a higher value reflects better clustering by rewarding compact clusters that are far apart from each other[11]. Together, these metrics provide a well-rounded assessment of clustering performance.

3 Results and Discussion

3.1 Cluster Assignments

The final K-Means model grouped the 20 countries into five clusters, resulting in four countries per cluster. Table 3 presents the cluster assignments and descriptive labels.

Cluster	Countries
0 - High Employment, Low AI Readiness	China, Germany, Spain, Turkey
1 - High Productivity, High Job Creation, High AI Readiness	Brazil, Japan, Netherlands, United States
2 - High Automation, Low Reskilling, High Job Displacement	France, Mexico, South Africa, South Korea
3 - Low Employment, Low Productivity	Australia, Italy, Singapore, United Kingdom
4 - High Reskilling, High AI Policy, Low Job Displacement	Canada, India, Russia, Sweden

Table 3. Cluster assignments and descriptive labels.

The five-cluster solution provided a more detailed grouping compared with the previous three-cluster version. The clusters showed differences in AI investment, automation rate, employment rate, productivity, reskilling investment, policy readiness, job displacement, job creation, and AI readiness score.

3.2 Standardized Cluster Profiles

Table 4 presents the standardized (z-score) mean values of each feature for each cluster. Positive values indicate a cluster is above the overall average for that feature. Negative values indicate below-average performance. This table is the primary basis for cluster interpretation.

Feature	Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4
AI_Investment_BillionUSD	0.51	0.81	0.43	-0.66	-1.09
Automation_Rate_Percent	-0.73	-0.31	1.38	0.40	-0.74
Employment_Rate_Percent	0.96	-0.47	0.39	-0.70	-0.19
Productivity_Index	-0.81	1.53	0.50	-0.97	-0.26
Reskilling_Investment_MillionUSD	-0.46	0.51	-0.91	-0.42	1.28
AI_Policy_Index	0.02	-0.42	-0.66	0.39	0.66
Job_Displacement_Million	-0.41	-0.07	1.09	0.53	-1.13
Job_Creation_Million	0.67	0.73	0.40	-0.77	-1.03
AI_Readiness_Score	-1.22	1.21	0.52	-0.22	-0.29

Table 4. Standard profile of each cluster across features.

To further enhance clarity and provide a clearer visual comparison across clusters, a heatmap representation of the same standardized values is presented in Figure 5. Unlike the tabular format, the heatmap allows for immediate visual identification of patterns, making it easier to observe relative strengths and weaknesses across clusters.

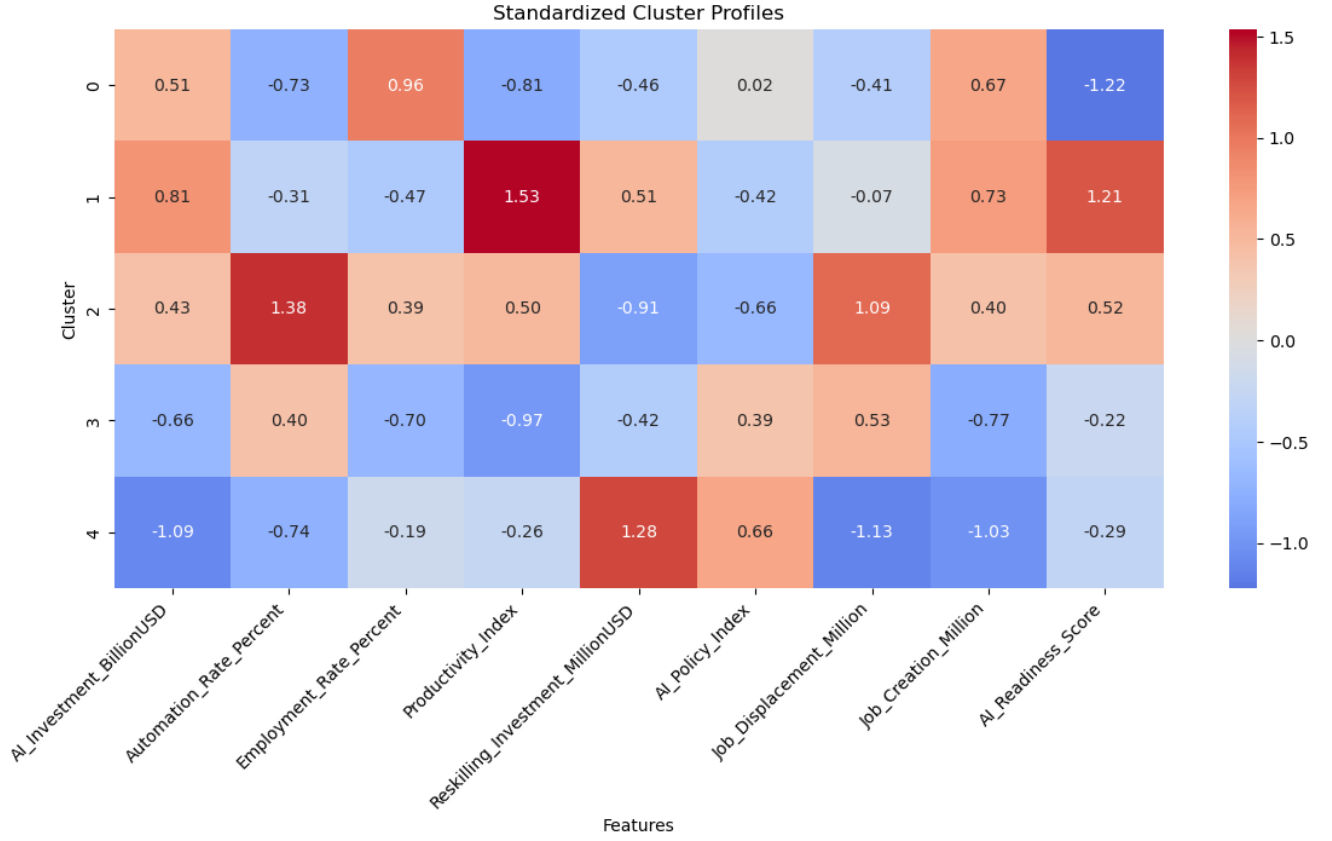


Fig. 5. Standardized feature profile heatmap for the five clusters.

Figure 5 summarizes the standardized feature profiles of the five clusters. Cluster 0 shows above-average employment rates (0.96) and AI investment levels (0.51) despite having low AI readiness (-1.22). Cluster 1 demonstrates high productivity (1.53) and AI readiness (1.21), while showing below-average employment rates (-0.47). Cluster 2 shows high automation rates (1.38) and job displacement (1.09), while having low reskilling investment (-0.91) and weak AI policy (-0.66). Cluster 3 has moderate feature values ranging from 0.53 to -0.97. It shows below-average AI investment (-0.66) and productivity (-0.97). Lastly, Cluster 4 displays high reskilling investment (1.28) while showing low AI investment (-1.09) and low job displacement (-1.13).

3.3 Cluster Quality: Silhouette Scores

Table 5 presents the mean, minimum, and maximum silhouette scores for each cluster. The overall silhouette score of 0.151 indicates weak to modest separation. Therefore, the five-cluster solution should be interpreted as an exploratory grouping rather than a strongly separated clustering structure.

Cluster	Average Silhouette Score	Minimum	Maximum
0	0.111	0.02	0.23
1	-0.032	-0.20	0.18
2	0.207	0.13	0.31
3	0.210	0.14	0.31
4	0.261	0.18	0.36

Table 5. Average silhouette score, minimum and maximum score per cluster for the five-cluster solution.

From the table, Cluster 4 obtained the highest average silhouette score (0.261), which indicates that the countries in that cluster are relatively well grouped and more clearly separated from other clusters.

Cluster 2 (0.207) and Cluster 3 (0.210) also demonstrate moderate cluster separation and cohesion. Cluster 0 shows lower average silhouette score (0.111), suggesting weaker separation between clusters and possible overlap with neighboring groups. Lastly, Cluster 1 has a negative average silhouette score (-0.032), suggesting some observations are positioned closer to the other clusters than their own cluster.

3.4 Cluster Quality: Davies-Bouldin Index and Dunn's Index

Table 6 shows the overall cluster validation metrics for the five clusters, including the Silhouette Score, Davies-Bouldin Index, and Dunn's Index.

Silhouette Score	Davies-Bouldin Index	Dunn's Index
0.151	1.277	0.453

Table 6. Overall cluster validation metrics for the five-cluster K-Means model.

Based on the results, the clustering model achieved an overall score of 0.151, indicating modest cluster separation with some overlap among clusters. The Davies-Bouldin Index of 1.277 suggests moderate cluster separation, meaning the clusters are not highly distinct but still show measurable differences in their profiles. Meanwhile, the Dunn's Index value of 0.453 indicates moderate inter-cluster separation relative to intra-cluster distance. Overall, the validation metrics suggest that the five clusters captured meaningful differences among countries, although some clusters exhibit overlapping characteristics.

3.5 Cluster Interpretation

Table 7 presents the countries assigned to each cluster and their corresponding interpretations.

Cluster	Countries	Interpretation
0	China, Germany, Spain, Turkey	Employment-strong, lower-readiness group
1	Brazil, Japan, Netherlands, United States	High-investment, high-readiness group
2	France, Mexico, South Africa, South Korea	Automation-heavy transition group
3	Australia, Italy, Singapore, United Kingdom	Mid-level automation with limited labor-market gains
4	Canada, India, Russia, Sweden	Policy- and reskilling-focused group

Table 7. Countries assigned to each cluster in the five-cluster K-Means model.

Cluster 0, which includes China, Germany, Spain, and Turkey, shows relatively strong employment but lower AI readiness compared with other groups. Although China and Germany are globally recognized for technological development, the standardized cluster profile indicates that these countries, along with the other members of the cluster, have relatively low AI readiness and reskilling. This suggests that stable employment does not necessarily correspond to higher AI readiness or productivity in the cluster. This cluster can be interpreted as an employment-strong but lower-readiness group.

Cluster 1, which includes Brazil, Japan, Netherlands, and United States, shows higher AI investment, productivity, job creation, and AI readiness. This cluster shows a pattern where higher AI investment appears alongside higher productivity, job creation, and AI readiness. This cluster can be interpreted as a high-investment and high-readiness group.

Cluster 2, which includes France, Mexico, South Korea, and South Africa, shows higher automation rates and higher job displacement. The low reskilling investment and low AI policy suggest that automation-related disruption may be more visible in this cluster, while reskilling and policy support are relatively weaker. This cluster can be interpreted as an automation-heavy transition group.

Cluster 3, which includes Australia, Italy, Singapore, and United Kingdom, shows moderate automation and moderate AI policy. However, productivity and job creation are low in this cluster. Its AI investment is also low compared to other clusters. This suggests that moderate policy readiness may not be enough to produce stronger productivity or job creation when AI investment remains comparatively low. This cluster can be interpreted as a mid-level automation group with limited labor-market gains.

Cluster 4, which includes Canada, Russia, India, and Sweden, shows stronger reskilling investment and AI policy index with lower job displacement and job creation. This pattern suggests that stronger policy and reskilling indicators may be related to lower job displacement in this cluster. The low job creation is likely associated with the low AI investment compared to other clusters. This cluster can be interpreted as a policy- and reskilling-focused group.

3.6 PCA Visualization

Figure 6 provides a two-dimensional visualization of the clustered countries after dimensionality reduction using PCA.

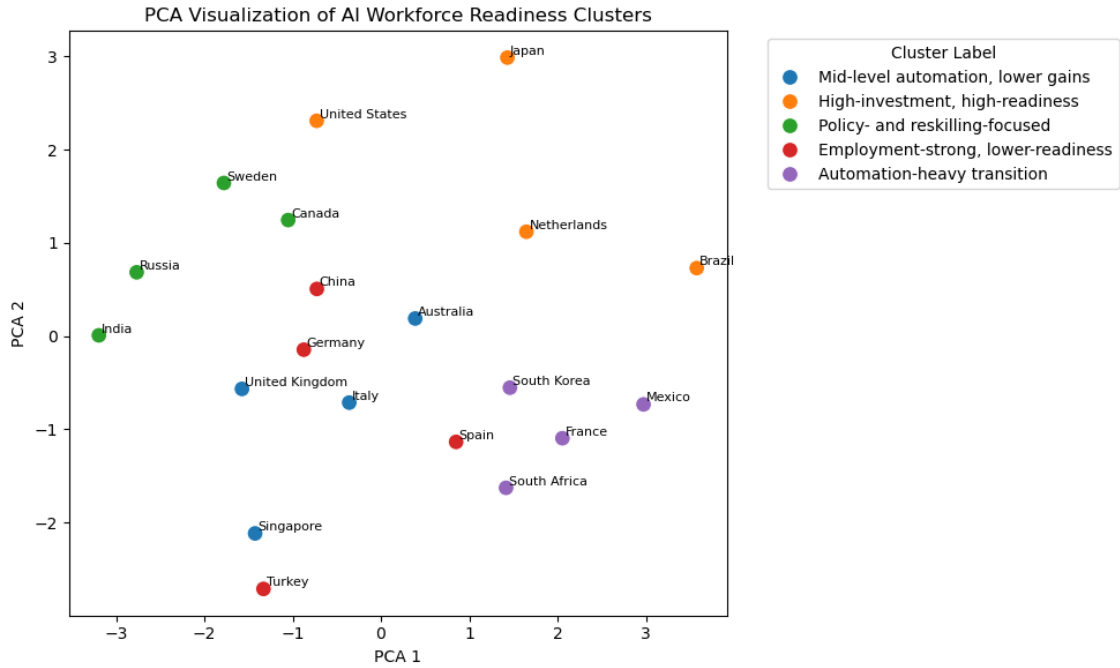


Fig. 6. PCA visualization of country clusters in two-dimensional space.

As discussed earlier, PCA was applied exclusively for visualization purposes. Countries such as Canada, Sweden, India, and Russia appear closer together in the plot. These countries share similar AI workforce and automation profiles as members of Cluster 4. In contrast, countries such as Japan, United States, Netherlands, and Brazil are farther apart from the other clusters. The visualization generally reflects the cluster boundaries identified by the model, although some overlap is visible. This overlap is consistent with the modest silhouette scores reported earlier.

3.7 World Map

Figure 7 presents the same cluster assignments geographically, making it easier to observe the distribution of country groups on the world map.

World Map of AI Workforce Readiness Clusters

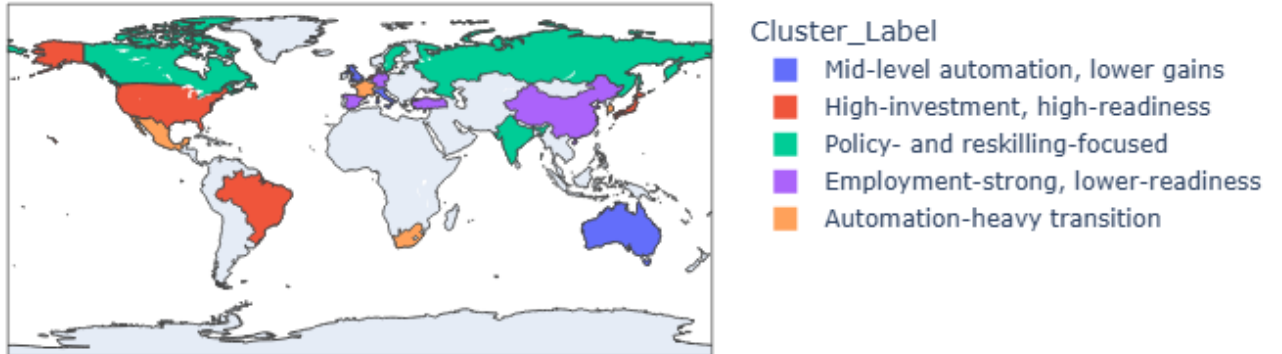


Fig. 7. World map visualization of countries colored according to their assigned cluster.

The map reveals that countries in the same cluster are not always geographically close to one another, suggesting that AI workforce readiness is driven more by investment and policy decisions than by regional location.

The results suggest that countries differ not only in their level of AI adoption but also in their approach to workforce readiness. Some clusters show stronger investment and productivity, while others show stronger policy and reskilling support. These findings are consistent with the study's objective of discovering meaningful country groupings based on AI workforce and automation indicators. The five-cluster solution reveals distinct profiles across the 20 countries, although the small cluster sizes should be considered a limitation.

The silhouette score analysis was useful in checking cluster separation. While smaller values of k may produce stronger silhouette scores, the five-cluster model was selected because it provides more detailed and interpretable groupings, consistent with the project feedback. As noted in the cluster quality analysis, the performance metrics indicate moderate cluster separation, which is expected given the small sample size of 20 countries.

4 Conclusion and Future Work

This project applied K-Means clustering to group countries based on AI workforce readiness and automation indicators from 2015 to 2025. The final model used five clusters and revealed different country profiles, including high-investment and high-readiness economies, automation-heavy transition economies, and policy- and reskilling-focused economies.

The findings suggest that AI workforce readiness is multidimensional. Countries may differ in investment, automation adoption, productivity, policy readiness, reskilling, and labor market outcomes. These results may help policymakers and researchers compare national AI workforce strategies and identify whether a country's main gap is investment, reskilling, policy support, or labor-market transition.

Future work may include using a larger dataset with more countries, analyzing yearly trends instead of only country-level averages, comparing K-Means with other clustering algorithms, and testing other feature combinations. Additional validation using real-world labor and AI policy data may also improve the reliability of the clustering results.

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